



DESHA EXAMINATION BOARD

2024 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

BIOLOGY

Subject Number: M164/I

Time Allowed: 2 hours

Tuesday 19th March, 2024

PAPER I

(100 marks)

THEORY

Instructions

1. This paper contains **10 pages**. Please check.
2. Before beginning, fill in your **Examination Number** at the top of each page of this question paper.
3. This paper contains sections **A** and **B**.
Section **A** has **ten** questions and
Section **B** has **three** questions
4. Answer all the 13 questions in the spaces provided. The maximum number of marks for each answer is indicated against each question
5. In the table provided on this page, tick against the question number you have answered.
6. You should hand in your question paper to the invigilator when you are called to stop writing.

Question number	Tick of answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Section A (70 Marks)

Answer **all** the questions in this section

1. a. Name any **two** adaptations of the spongy mesophyll for their function in a leaf.

(2 marks)

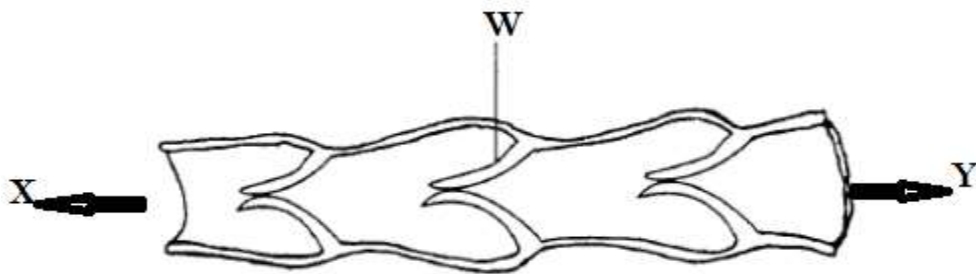
- b. Mention **two** mineral elements in photosynthesis for chlorophyll formation.

(2 marks)

- c. Describe how transpiration stream occurs in plants.

(4 marks)

2. **Figure 1** is a diagram showing a vessel of the circulatory system. Use it to answer questions that follow;



- a. Name any **two** vessels in the human body which have structure **W**.

(2 marks)

- b. What could be the possible direction of blood flow on the diagram?

(1 mark)

3. a. Define the term Biotechnology;

(1 mark)

b. State any **two** benefits associated with biotechnology that can lead to food security.

(2 marks)

c. Explain how insulin is produced.

(4 marks)

4. **Figure 2** is the diagram showing positions of endocrine glands in humans. Use it to answer the questions that follow.



a. Name the glands labelled **Z** and **W**

Z _____

W _____

(2 marks)

b. Explain any **one** way in which gland labelled **Y** regulates blood sugar level.

(2 marks)

c. State any **two** differences between endocrine system and nervous system.

(4 marks)

5 a. State the **two** types of cancer.

(2 marks)

b. Fill the table below by relating the following diseases; common cold, ring worm and cholera according to their given causative agents.

Viral diseases	Fungal diseases	Bacterial diseases

(3 marks)

6 a. Which human hormone affects water reabsorption by the kidney tubules?

(1 mark)

b. Give any **two** accessory glands in a male reproductive system.

(2 marks)

c. Write down **three** advantages of exclusive breast feeding.

(3 marks)

d. Explain any **two** adaptations of the placenta to its functions.

(4 marks)

7. a. State any **two** problems associated with the human digestive system.

(2 marks)

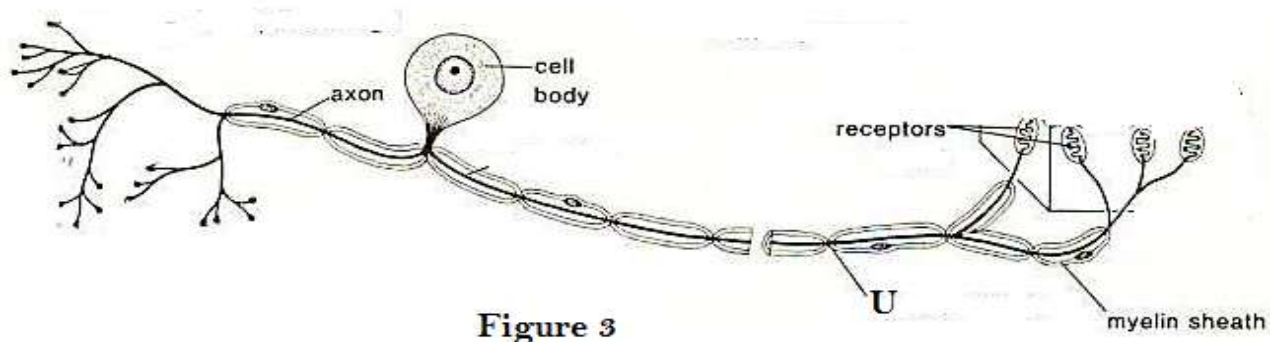
b. State any **two** areas where physical digestion takes place along the human digestive system.

(2 marks)

c. Explain any **two** adaptations of the villi for food absorption.

(4 marks)

8. **Figure 3** is diagram of a neurone. Use it for questions that follow;



a. Identify the neurone

(1 mark)

b. Give any **two** features that enabled you to come up with the answer above.

(2 marks)

9 a. Explain any **two** characteristics of respiratory surfaces for gaseous exchange.

(4 Marks)

b. State any **two** similarities between a dialysis machine and the tubule of the nephron.

(2 Marks)

c. Explain any **two** adaptations of birds for their locomotion.

(4 Marks)

10. a. Give any **two** example of discontinuous variation.

(2 marks)

b. A potted plant when placed horizontally grows away from the pull of gravity.
Explain how this response happens.

(2 marks)

c. In rats a gene for coat colour has two alleles. There is a dominant allele **G** for grey colour and a recessive allele **g** for white colour.

Complete the table below using given information.

Colour of rats	Genotype of rats
Heterozygous grey	
Homozygous white	

(2 marks)

d. Explain according to Mendel experiments how a pure plant breed of any trait could be obtained.

(2 Marks)

12. Discuss any **five** evidences of evolution..

[illegible]

(10 marks)

13. Explain any **five** effects of exercises on breathing.

[illegible]

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 10 printed pages